

Checklist for the functionality of the products

The list of features that will be checked during the final demonstration are listed below:

IPOS-SA sub-system

IPOS-SA-ACC package

- creating/changing/deleting a user account
 - ability to create new user accounts (login/password) and assign roles (access privileges)
 - users should be allowed to do only things defined by their access privileges and not allowed to do things which exceed their privileges.
 - change roles (promote/demote) users to different level of access.
 - This can be done via deleting an existing user account and creating it with different privileges.
- creating/changing/deleting a merchant account
- setting up a credit limit and a discount plan for a merchant account
- maintaining account state (from “normal” to “suspended” and eventually to “in default”) including reactivating accounts “in default” to “normal”

IPOS-SA-CAT package

- adding/deleting/modifying a product in the catalogue
 - setting/changing minimum stock availability.
- search in the catalogue by product ID, keyword, etc.
- adding new stock (i.e. increasing stock availability)
- warnings for stock below the set lower bound (e.g. when a stock item is open for modification, or when a user logs in)

IPOS-SA-ORD package

- accepting new orders (submitted from IPOS CA),
 - generating invoices and storing them on SA.
- managing the order status (from “accepted” to “ready to dispatch”, then to “dispatched”, then to delivered).
- monitoring current balance of a merchant.
- Recording payments from merchants using various forms of payment (e.g. bank transfers).
 - Recording payments may change the merchant account status from “suspended” to “normal” provided the outstanding balance from previous month has been cleared.
- observing the list of orders taken but not completed.

IPOS-SA-RPT package

- ability to generate all reports required with an option to print them.
 - generating a file (e.g. a pdf file) will be considered equivalent to printing.
- generating reminders to merchant debtors (shown on CA screen) every time a merchant debtor tries to log in.

IPOS-CA sub-system

IPOS-CA-ORD package

- placing new orders with SA
- checking progress with SA order
- search of SA orders and viewing invoices.
- Check status of own merchant account.
- (optional) record merchant account details on CA for ease of access to SA by different users of IPOS - CA.

IPOS-CA-USER package

- create/change/delete an IPOS-CA user account with different privileges.

IPOS-CA-CUST package

- create/change/delete an IPOS-CA customer account.
 - set up/modify a credit limit.
 - set up/modify a discount plan for an existing account holder (customer).
- generate reminders to debtors (1st and 2nd reminders)
- maintain customer account (from “normal” to “suspended” and then to “in default”) including reactivation of an account “in default” by an authorised user, provided the debt from the previous month has been cleared.

IPOS-CA-Templates package

- Ability to create/modify reminder templates (1st and 2nd reminder).
 - Make sure that the templates are used in any future reminders.

IPOS-CA-Stock package

- dealing with local stock:
 - adding new stock (a new item and increasing the quantity of an existing stock item)
 - dealing with order deliveries (when an IPOS-SA order is delivered)¹
 - listing (on request) all items, which are below their respective lower stock bounds.

IPOS-CA-Sales package

- process purchases by customers, produce retail invoices.
- take payments from customers using various forms of payment (including cash and credit card)
- maintain orders received via the PU portal (order status will be changed from “accepted” to “ready for shipment”, then to “shipped” and finally “delivered”).

IPOS-CA-RPT package

- ability to generate all reports required with option to print them.
 - generating a file (e.g. a pdf file) will be considered equivalent to printing.

IPOS-PU sub-system

IPOS-PU-Members package

- application for non-commercial account (membership)
- application for commercial account (membership)
- login/ by non-commercial members
- non-commercial customer can view their purchase history.
- ability by non-commercial members to track their orders (see the state of orders).

IPOS-PU-Sales package

- show IPOS - CA catalogue.
 - show a list of links to active promotion campaigns (provided there is at least one active campaign at the time a customer is scanning the on-line catalogue)
 - clicking the link of an active promotion campaign will show details on promotion campaign.
- maintain a shopping cart for customers (and non-commercial members)
 - apply discounts to products included in active promotion campaigns.
 - maintain all counters envisaged in the initial statement of requirements (when links/product items are selected/added to cart/paid for).
- checkout
 - accounting for non-commercial customer getting a 10% discount from every tenth order.
 - Collect delivery address.
 - Collect online payment.
 - Send the customer a sale confirmation with an order tracking link.

¹ An assumption can be made that all orders placed with IPOS-SA are delivered as placed (i.e. the delivered items and quantities are as specified in the order).

IPOS-PU-Comms package

- (optional) Configure access to SMTP server.
- (optional) Configure access to an online payment processor.
- Sending mails on behalf of IPOS-SA
- Processing credit card payment requests on behalf of IPOS-CA (card payments are processed via the online payment processor)

IPOS-PU-PRM package

- Create/modify/delete/terminate early promotion campaigns.
 - Setting period of campaign
 - Including items from IPOS-CA catalogue
 - Setting discount rates

IPOS-PU-RPT package

- Create all reports envisaged in the initial requirements documents with option to print them.
 - generating a file (e.g. a pdf file) will be considered equivalent to printing.

G: General functionality

- Sensible search
- Cascaded deletes/updates
- GUI – appearance of the forms (intuitive, easy to use) and consistency between the forms.

Sample Data

The data provided below should be used to populate the databases of IPOS-SA, IPOS-CA, and IPOS – PU subsystems. Prior to populating the databases, you **MUST** remove all the data from the database of the respective subsystem, i.e. leave all tables blank, with a single exception – a single (default) login which will allow you to create the required login usernames/passwords.

IPOS_SA subsystem

Logins

The following user account must be active with IPOS-SA:

Username	Password	role
Sysdba	London weighting	Administrator
manager	Get it done	Director of Operations
accountant	Count money	Senior accountant
clerk	Paperwork	Accountant
warehouse1	Get a beer	Warehouse employee
warehouse2	Lot smell	Warehouse employee
delivery	Too dark	Delivery department employee

Account details of merchants registered with InfoPharma

Account Holder Name: CityPharmacy
Account No: ACC0001
Contact Name: Prof David Rhind
Address: Northampton Square, London EC1V 0HB
Phone: 0207 040 8000
Credit Limit: £10,000
Agreed Discount: Fixed
Discount Rate: 3%
login: city
password: northampton

Account Holder Name: Cosymed Ltd
Account No: ACC0002
Contact Name: Mr Alex Wright
Address: 25, Bond Street, London WC1V 8LS
Phone: 0207 321 8001
Credit Limit: £5,000
Agreed Discount: Variable (on volume per month)
Discount Rate:
Volume Discount Rate
< £1000 : 0 %
£1000 - £2000: 1%
£2000+ : 2 %
login: cosymed
password: bondstreet

Account Holder Name: HelloPharmacy
Account No: ACC0003
Contact Name: Mr Bruno Wright
Address: 12, Bond Street, London WC1V 9NS
Phone: 0207 321 8002
Credit Limit: £5,000
Agreed Discount: Variable (on volume per month)
Discount Rate:
Volume Discount Rate
< £1000 : 0 %
£1000 - £2000: 1%
£2000+: 3 %
login: hello
password: there

InfoPharma catalogue

Item ID	Description	Package Type	Unit	Units in a pack	Package Cost, £	Availability, packs	Stock limit, packs
100 00001	Paracetamol	box	Caps	20	0.10	10,345	300
100 00002	Aspirin	box	Caps	20	0.50	12, 453	500
100 00003	Analgin	box	Caps	10	1.20	4,235	200
100 00004	Celebrex, caps 100 mg	box	Caps	10	10.00	3,420	200
100 00005	Celebrex, caps 200 mg	box	caps	10	18.50	1,450	150
100 00006	Retin-A Tretin, 30 g	box	caps	20	25.00	2,013	200
100 00007	Lipitor TB, 20 mg	box	caps	30	15.50	1,562	200
100 00008	Claritin CR, 60g	box	caps	20	19.50	2,540	200
200 00004	Iodine tincture	bottle	ml	100	0.30	2,2134	200
200 00005	Rhynol	bottle	ml	200	2.50	1,908	300
300 00001	Ospen	box	caps	20	10.50	809	200
300 00002	Amopen	box	caps	30	15.00	1340	300
400 00001	Vitamin C	box	caps	30	1.20	3,258	300
400 00002	Vitamin B12	box	caps	30	1.30	2,673	300

IPOS_CA subsystem

The following *user* account must be active with the version of IPOS-CA deployed at **Cosymed Ltd.**:

Username	Password	Role
sysdba	masterkey	Administrator
manager	Get it done	Director of Operations/Manager
accountant	Count_money	Senior accountant
clerk	Paperwork	Accountant

Customers registered as account holders with Cosymed Ltd.

Account Holder Name: Ms Eva Bauyer
Account No: ACC0001
Contact Name: Ms Eva Bauyer
Address: 1, Liverpool street, London EC2V 8NS
Phone: 0207 321 8001
Credit Limit: £500
Agreed Discount: Fixed: 3%

Account Holder Name: Mr Glynne Morrison
Account No: ACC0002
Contact Name: Ms Glynne Morisson
Address: 1, Liverpool street, London EC2V 8NS
Phone: 0207 321 8001
Credit Limit: £500
Agreed Discount: Variable (on volume per month)
Discount Rate:

Volume	Discount Rate
< £100 :	0 %
£100 - £300:	1%
£300+ :	2 %

Stock availability at Cosymed Ltd

Item ID	Description	Package Type	Unit	Units in a pack	Package Cost, £	Availability, packs	Stock limit, packs
100 00001	Paracetamol	Box	Caps	20	0.10	121	10
100 00002	Aspirin	Box	Caps	20	0.50	201	15
100 00003	Analgin	Box	Caps	10	1.20	25	10
100 00004	Celebrex, caps 100 mg	Box	Caps	10	10.00	43	10
100 00005	Celebrex, caps 200 mg	Box	Caps	10	18.50	35	5
100 00006	Retin-A Tretin, 30 g	Box	Caps	20	25.00	28	10
100 00007	Lipitor TB, 20 mg	Box	Caps	30	15.50	10	10
100 00008	Claritin CR, 60g	Box	Caps	20	19.50	21	10
200 00004	Iodine tincture	Bottle	MI	100	0.30	35	10
200 00005	Rhynol	Bottle	MI	200	2.50	14	15
300 00001	Ospen	Box	Caps	20	10.50	78	10
300 00002	Amopen	Box	Caps	30	15.00	90	15
400 00001	Vitamin C	Box	Caps	30	1.20	22	15
400 00002	Vitamin B12	Box	Caps	30	1.30	43	15

Retail prices are calculated with 100% mark up to the unit cost. 0.0% VAT is applied to the retail price.

IPOS_PU subsystem

The following *user* accounts must be active with the version of IPOS-PU linked to merchant **Cosymed Ltd**:

Username	Password	Role
sysdba	masterkey	Administrator
manager	GetPU it done	PU-Admin

The following membership applications were made via IPOS-PU (Cosymed Ltd):

Non-commercial Member

Account: PU0001
Email address: cool@example.com²
password: 12ss_56_SS

Non-commercial Member

Account No: PU0002
Email address: cool1@example.com³
password: 34pp_78_LL

Commercial Member

Account No: PU0003
Company Name: Pond Pharmacy
Address: Chislehurst
25, High Street
BR7 5BN
Company House Registration: UK10003429CompH
Email address: pondPharma@example.com⁴

² You can amend this email to a valid email address, so that emails can be sent to it.

³ You can amend this email to a valid email address, so that emails can be sent to it.

⁴ You can amend this email to a valid email address, so that emails can be sent to it.

Scenarios

1. On 20 February 2026 CityPharmacy, placed the following order with IPOS-SA:

Item ID	Description	Quantity	Unit Cost, £	Total, £
100 00001	Paracetamol	10	0.10	1.00
100 00003	Analgin	20	1.20	24.00
200 00004	Iodine tincture	20	0.30	3.60
200 00005	Rhynol	10	2.50	25.00
300 00001	Ospen	10	10.50	105.00
300 00002	Amopen	20	15.00	300.00
400 00001	Vitamin C	20	1.20	24.00
400 00002	Vitamin B12	20	1.30	26.00
Grand Total:				508.60

The goods were delivered on 23 February 2026, 15:00 to the merchant's premises by the InfoPharma's own courier service.

2. On 25 February 2026, the merchant Cosymed Ltd, placed the following order with IPOS-SA:

Item ID	Description	Quantity	Unit Cost, £	Total, £
100 00001	Paracetamol	10	0.10	1.00
100 00003	Analgin	20	1.20	24.00
200 00005	Rhynol	10	2.50	25.00
300 00002	Amopen	20	15.00	300.00
400 00002	Vitamin B12	20	1.30	26.00
Grand Total:				376.00

The goods were delivered on 26 February, 17:00 to the merchant's premises by the DHL.

3. On 25 February 2026, the merchant HelloPharmacy placed the following order with IPOS - SA:

Item ID	Description	Quantity	Unit Cost, £	Total, £
100 00003	Analgin	20	1.20	24.00
200 00004	Iodine tincture	20	0.30	3.60
300 00001	Ospen	3	10.50	31.50
300 00002	Amopen	10	15.00	150.00
400 00001	Vitamin C	20	1.20	24.00
400 00002	Vitamin B12	20	1.30	26.00
Grand Total:				259.10

The goods were delivered on 27 February 2026, 10:00 to the merchant's premises by the DHL.

4. On 10 March 2026, the merchant Cosymed Ltd placed the following order with IPOS - SA:

Item ID	Description	Quantity	Unit Cost, £	Total, £
200 00005	Rhynol	10	2.50	25.00
300 00001	Ospen	10	10.50	105.00
300 00002	Amopen	20	15.00	300.00
Grand Total:				430.00

The goods were delivered on 12 March, 11:00 to the merchant's premises by the InfoPharma's own courier service.

5. On 25 March 2026, the merchant HelloPharmacy placed the following order with IPOS - SA:

Item ID	Description	Quantity	Unit Cost, £	Total, £
100 00003	Analgin	20	1.20	24.00
100 00004	Celebrex, caps 100 mg	5	10.00	50.00
100 00005	Celebrex, caps 200 mg	5	18.50	92.50
100 00006	Retin-A Tretin, 30 g	5	25.00	125.00
100 00007	Lipitor TB, 20 mg	10	15.50	155.00
300 00001	Ospen	10	10.50	105.00
300 00002	Amopen	20	15.00	300.00
400 00002	Vitamin B12	20	1.30	26.00
Grand Total:				877.50

The goods were delivered on 27 March, 10:00 to the merchant's premises by the InfoPharma's own courier service.

6. On 1 April 2026, the merchant HelloPharmacy placed the following order with IPOS - SA:

Item ID	Description	Quantity	Unit Cost, £	Total, £
100 00003	Analgin	20	1.20	24.00
100 00004	Celebrex, caps 100 mg	5	10.00	50.00
100 00005	Celebrex, caps 200 mg	5	18.50	92.50
100 00006	Retin-A Tretin, 30 g	5	25.00	125.00
100 00007	Lipitor TB, 20 mg	10	15.50	155.00
300 00001	Ospen	10	10.50	105.00
400 00002	Vitamin B12	20	1.30	26.00
Grand Total:				577.50

The goods were delivered on 3 April 2026, 10:00 to the merchant's premises by the InfoPharma's own courier service.

7. HelloPharmacy have made no payments since 5 March 2026 when they cleared their balance.

8. CityPharmacy have made a full payment on 15 March 2026 and cleared their balance using a bank transfer to the account of InfoPharma.

9. Cosymed Ltd have made a full payment on 15 March 2026 and cleared their balance using company's credit card.

10. On 1 March 2026 Eva Bauyer bought the following goods from Cosymed Ltd:

Ospen	1 Boxes
Vitamin C	2 Boxes
Amopen	2 Boxes
Vitamin B12	2 Boxes

11. On 3 March 2026, the following purchases of goods were made from the pharmacy shops of Cosymed Ltd (i.e. via the IPOS-CA subsystem) by several cash customers:

a)

Aspirin	2 Boxes
Analgin	3 Boxes

For goods, the customer paid cash.

b)

Celebrex, caps 100 mg	2 Boxes
Retin-A Tretin, 30 g	2 Boxes

Paid with credit card Visa.

c)

Lipitor TB, 20 mg	1 Box
Claritin CR, 60g	1 Box

Paid cash.

d)

Celebrex, caps 200 mg	1 Box
Iodine tincture	2 Bottles
Rhynol	2 Bottles

Paid with cash.

e)

Ospen	2 Boxes
Vitamin C	2 Boxes

Paid with debit card.

f)

Amopen	3 Boxes
Vitamin B12	2 Boxes

Paid cash.

12. On 5 March 2026 Glynne Morisson bought from a Cosymed Ltd Pharmacy Shop Ltd the following goods:

Aspirin	2 Boxes
Analgin	3 Boxes
Celebrex, caps 100 mg	2 Boxes
Retin-A Tretin, 30 g	2 Boxes

The cost of goods was added to Glynne's account with Cosymed Ltd.

13. On 1 April Eva Bauyer bought the following goods from a Cosymed Pharmacy shop:

Ospen	1 Boxes
Analgin	3 Boxes
Celebrex, caps 100 mg	2 Boxes
Vitamin B12	2 Boxes

The cost of goods was added to Eva's account with Cosymed Ltd.

14. Glynne Morisson paid on 29 March the full balance with a credit card.

15. The last payment from Eva Bauyer was received on 30 February.

16. On the 15 April reminding letters were generated and sent to all debtors. The same must be done on 29 April.

17. On the 15 March 2026, the Admin – PU user of IPOS-PU created a promotion campaign as follows:

Name:	March Promotion
Valid from:	15 March 2026
Valid to:	20 April 2026
Item	Discount rate

Aspirin	5%
Analgin	10%
Celebrex, caps 100 mg	10%
Retin-A Tretin, 30 g	20%

18. On the 30 March 2026, the Admin – PU user of IPOS-PU created another promotion campaign as follows:

Name:	April Promotion
Valid from:	5 April 2026
Valid to:	10 April 2026
Item	Discount rate

Ospen	20%
Vitamin C	10%

19. On the 20 of March 2026 Peter Popov scanned the Cosymed Ltd online catalogue (via the PU subsystem), clicked on the link of "March Promotion". Then from the list of the promotion campaigns, he added two items to the shopping cart:

- Aspirin - 1 box
- Retin-A Tretin, 30 g – 1 box.

He then paid using his AmEx credit card: 0000 000000 0000 0001, security code 3245, valid until 30/08/2030. The payment has been successful. Peter was sent an email confirming the sale and with a link to track the delivery.

20. Between 1st and 20th of March 2026 the non-commercial customer PU0001 (see login details above) has made 8 online purchases of various goods (details of these have not been stored, but the number of purchases (8) has been stored by the PU). On the 8th of April 2026 PU0001 logged in again and checked the active promotion campaigns. Then they added Ospen – 1 box to the shopping cart. They checked out and paid for the goods using their Mastercard 0001 0002 0003 0005 with security code 032 and expiry date 9/2028. The payment transaction has been successful and PU0001 received an email confirming the purchase with an order tracking link included in the email.